

ME101: Engineering Mechanics, 2023-24

IIT Guwahati

Tutorial – 10 (Solution)

Tutorial Date: 8th April 2024 (7:55 am – 8:50 am)

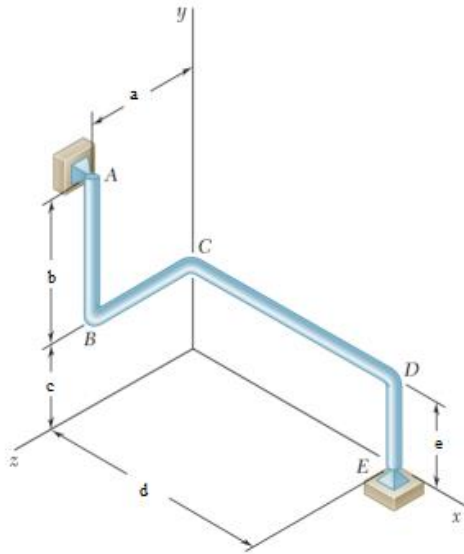


Figure 1: Problem 1

Problem 1:

The bent rod ABCDE rotates about a line joining Points A and E with a constant angular velocity of 20 rad/s. Knowing that the rotation is clockwise as viewed from E, determine the velocity and acceleration of corner C. Assume the dimensions are $a = 40\text{mm}$, $b = 50\text{mm}$, $c = 30\text{mm}$, $d = 100\text{mm}$, $e = 30\text{mm}$. [10 Marks]

Solution:

Here, $a = 40\text{mm}$, $b = 50\text{mm}$, $c = 30\text{mm}$, $d = 100\text{mm}$, $e = 30\text{mm}$.

$$EA^2 = 0.1^2 + 0.08^2 + 0.04^2, \quad EA = 0.134 \text{ m}$$

$$\mathbf{r}_{C/E} = -(0.1 \text{ m})\mathbf{i} + (0.03\text{m})\mathbf{j}$$

As rotation is clockwise, viewed from E, using right hand rule, $\vec{\omega}$ is acting along vector from E to A.

$$\vec{EA} = -(0.1 \text{ m})\mathbf{i} + (0.08\text{m})\mathbf{j} + (0.04\text{m})\mathbf{k}$$

Unit vector from E to A,

$$\lambda_{EA} = \frac{\vec{EA}}{EA} = \frac{1}{0.134}(-0.1\mathbf{i} + 0.08\mathbf{j} + 0.04\mathbf{k}) = \frac{1}{67}(-50\mathbf{i} + 40\mathbf{j} + 20\mathbf{k})$$

$$\boldsymbol{\omega} = \omega_{AE}\lambda_{EA} = (20 \text{ rad/s})\frac{1}{67}(-50\mathbf{i} + 40\mathbf{j} + 20\mathbf{k})$$

$$\boldsymbol{\omega} = (-14.925 \text{ rad/s})\mathbf{i} + (11.94 \text{ rad/s})\mathbf{j} + (5.97 \text{ rad/s})\mathbf{k}$$

$$\mathbf{v}_C = \boldsymbol{\omega} \times \mathbf{r}_{C/E} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -14.925 & 11.94 & 5.97 \\ -0.1 & 0.03 & 0 \end{vmatrix} = -0.179\mathbf{i} - 0.597\mathbf{j} + 0.746\mathbf{k}$$

$$\mathbf{v}_C = (-0.179 \text{ m/s})\mathbf{i} - (0.597 \text{ m/s})\mathbf{j} + (0.746 \text{ m/s})\mathbf{k} \quad [5 \text{ Marks}]$$

You can take any vector from any point on axis of rotation ($\vec{EA}$) to point C.

Let us repeat the calculation using $\mathbf{r}_{C/A} = -0.04\mathbf{k} - 0.05\mathbf{j}$.

$$\text{then, } \mathbf{v}_c = \boldsymbol{\omega} \times \mathbf{r}_{C/A} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -14.925 & 11.94 & 5.97 \\ 0 & -0.05 & -0.04 \end{vmatrix} = -0.179\mathbf{i} - 0.597\mathbf{j} + 0.746\mathbf{k}$$

Here $\vec{\omega}$ is a constant, so $\vec{\alpha} = 0$.

$$\mathbf{a}_c = \boldsymbol{\alpha}_{AC} \times \mathbf{r}_{C/E} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{C/E}) = \boldsymbol{\alpha}_{AC} \times \mathbf{r}_{C/E} + \boldsymbol{\omega} \times \mathbf{v}_c$$

$$\mathbf{a}_c = 0 + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -14.925 & 11.94 & 5.97 \\ -0.179 & -0.597 & 0.746 \end{vmatrix} = 12.474\mathbf{i} + 10.068\mathbf{j} + 11.048\mathbf{k}$$

$$\mathbf{a}_c = (12.474 \text{ m/s}^2)\mathbf{i} + (10.068 \text{ m/s}^2)\mathbf{j} + (11.048 \text{ m/s}^2)\mathbf{k} \quad [5 \text{ Marks}]$$

Problem 2:

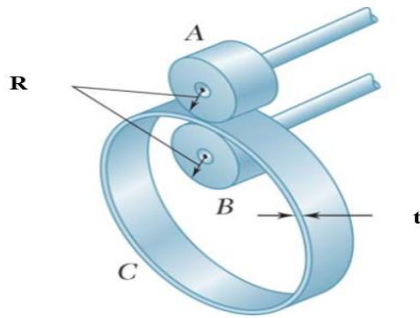


Figure 2: Problem 2

Ring C has an inside radius of 55 mm and an outside radius of 60 mm, here t is 5 mm and is positioned between two wheels A and B, each of 24-mm (R) outside radius. Knowing that wheel A rotates with the clockwise angular velocity of 300 rpm which reduces at the rate of 20 rad/s^2 and that no slipping occurs, determine (a) the angular velocity of the ring C and wheel B, (b) the acceleration of the points on A and B that are in contact with C. [10

Marks]

Solution:

$$\omega_A = 300 \text{ rpm} \left(\frac{2\pi}{60} \right) = 31.416 \text{ rad/s}; \quad \vec{\omega}_A = -31.416 \hat{\mathbf{k}} \text{ rad/s}$$

$$R = r_A = r_B = 24 \text{ mm}; \quad t = 5 \text{ mm}; \quad r_1 = 60 \text{ mm}; \quad r_2 = 55 \text{ mm}$$

Velocities :

Point 1 - Point of contact between A and C

$$v_1 = r_A \omega_A = r_1 \omega_C; \quad \vec{v}_1 = -v_1 \hat{\mathbf{i}}$$

$$\omega_C = \frac{r_A}{r_1} \omega_A = \frac{24 \text{ mm}}{60 \text{ mm}} (300 \text{ rpm}) = 120 \text{ rpm}; \quad \vec{\omega}_C = 120 \hat{\mathbf{k}} \text{ rpm} \quad [2 \text{ Marks}]$$

Point 2 - Point of contact between B and C

$$v_2 = r_B \omega_B = r_2 \omega_C; \quad \vec{v}_2 = -v_2 \hat{\mathbf{i}}$$

Here, wheel B and ring C rotate in same direction i.e. counter-clockwise

$$\omega_B = \frac{r_2}{r_B} \omega_C = \frac{r_2}{r_B} \left(\frac{r_A}{r_1} \right) \omega_A = \frac{55 \text{ mm}}{24 \text{ mm}} \left(\frac{24 \text{ mm}}{60 \text{ mm}} \right) (300 \text{ rpm}) = 275 \text{ rpm}; \quad \vec{\omega}_B = 275 \hat{\mathbf{k}} \text{ rpm}$$

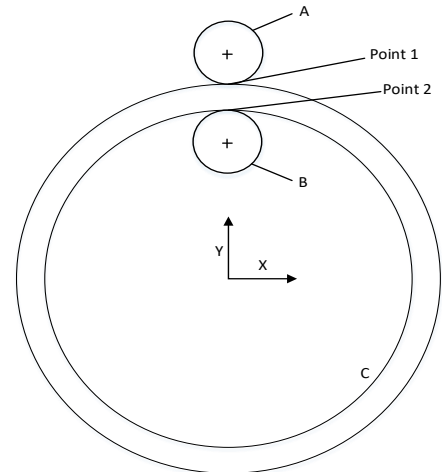


Figure 3

[2 Marks]

$$\vec{\omega}_A = -34.416 \hat{k} \text{ rad/s}; \vec{\alpha}_A = 20 \hat{k} \text{ rad/s}^2 \text{ [Here } \vec{\alpha}_A \text{ is in opposite direction as } \vec{\omega}_A \text{ is reducing]}$$

$$(\vec{a}_1)_t = \vec{\alpha}_A \times \vec{r}_A = 20 \hat{k} \times (-0.024 \hat{j}) = 0.48 \hat{i}; (a_1)_t = \alpha_A r_A = \alpha_c r_1$$

$$\text{Let, } \vec{\alpha}_c = \alpha_c \hat{k}, \vec{r}_1 = 0.06 \hat{j}, \therefore (\vec{a}_1)_t = \vec{\alpha}_c \times \vec{r}_1 \text{ or, } 0.48 \hat{i} = -0.06 \alpha_c \hat{i}; \alpha_c = 8 \text{ rad/s}^2$$

$$\therefore \vec{\alpha}_c = -8 \hat{k}$$

$$(a_2)_t = \alpha_c r_2 = \alpha_B r_B;$$

$$(\vec{a}_2)_t = \vec{\alpha}_c \times \vec{r}_2 = -8 \hat{k} \times (0.055 \hat{j}) = 0.44 \hat{i}$$

$$\text{Let, } \vec{\alpha}_B = \alpha_B \hat{k}$$

$$\therefore (\vec{a}_2)_t = \vec{\alpha}_B \times \vec{r}_B; \text{Or, } 0.44 \hat{i} = \alpha_B \hat{k} \times 0.024 \hat{j} = -0.024 \alpha_B \hat{i}; \alpha_B = -18.33 \text{ rad/s}^2$$

$$\therefore \vec{\alpha}_B = -18.33 \hat{k}$$

$$\vec{a}_A = \vec{\alpha}_A \times \vec{r}_A + \vec{\omega}_A \times \vec{\omega}_A \times \vec{r}_A = 20 \hat{k} \times (-0.024 \hat{j}) + (\omega_A^2 \times 0.024) \hat{j} = 0.48 \hat{i} + 23.68 \hat{j} \quad [3 \text{ Marks}]$$

$$\begin{aligned} (\vec{a}_2)_B &= \text{Acceleration of point 2 on B which is in contact with C} = \vec{\alpha}_B \times \vec{r}_B + \vec{\omega}_B \times \vec{\omega}_B \times \vec{r}_B \\ &= -18.33 \hat{k} \times (0.024 \hat{j}) - (\omega_B^2 \times 0.024) \hat{j} = 0.44 \hat{i} - 19.91 \hat{j} \text{ m/s}^2 \quad [3 \text{ Marks}] \end{aligned}$$

Problem 3:

Activation of the hydraulic cylinder causes OB to elongate at the constant rate of 0.5 m/s . Determine both tangential and normal acceleration of point A in its circular path around C for the instant when $\theta = 60^\circ$ and for $L_1 = 375 \text{ mm}, L_2 = 250 \text{ mm}, L_3 = 500 \text{ mm}, L_4 = 500 \text{ mm}$. [10 Marks]

Solution:

For $\triangle OBC$, Let, $OB = x \text{ m}$

$$x = [0.5^2 + 0.5^2 - 2 \times 0.5^2 \times \cos \theta]^{\frac{1}{2}}$$

$$= [2 \times 0.5^2 (1 - \cos \theta)]^{\frac{1}{2}} = 0.5 \times \sqrt{2} \times \sqrt{1 - \cos \theta}$$

$$\Rightarrow x^2 = 2 \times 0.5^2 (1 - \cos \theta)$$

$$\Rightarrow 2x \cdot \dot{x} = 2 \times 0.5^2 \times \sin \theta \dot{\theta}$$

$$\Rightarrow \dot{\theta} = (x \cdot \dot{x}) / (0.5^2 \times \sin \theta)$$

$$\text{When } \theta = 60^\circ, x = 0.5 \times \sqrt{2} \times \frac{1}{\sqrt{2}} = 0.5$$

$$\therefore \dot{\theta} = \frac{0.5 \times 0.5}{0.5^2 \times \sin 60^\circ} = 1.1547 \text{ rad/s} = \omega$$

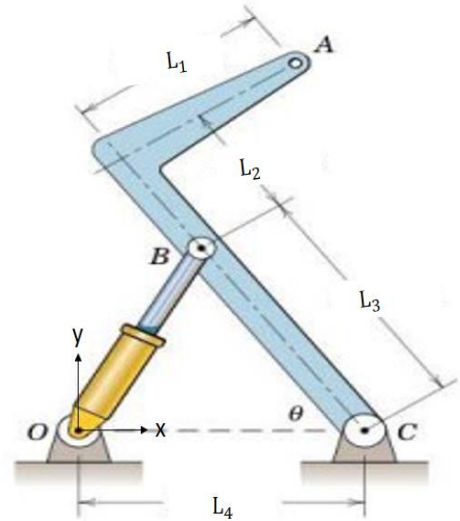


Figure 4: Problem 3

As $\dot{\theta}$ is positive $\Rightarrow \theta$ is increasing

$$\therefore \vec{\omega} = -1.1547 \hat{k} \text{ rad/s [Clockwise]} \quad [3 \text{ Marks}]$$

Now, $x\dot{x} = 0.5^2 \sin \theta \dot{\theta}$

$$\Rightarrow x\ddot{x} + \dot{x}^2 = 0.5^2 \cos \theta \dot{\theta}^2 + 0.5^2 \sin \theta \ddot{\theta}$$

$$\Rightarrow 0 + 0.5^2 = 0.5^2 \times \cos 60^\circ \times 1.1547^2 + 0.5^2 \times \ddot{\theta} \sin 60^\circ$$

$$\Rightarrow \ddot{\theta} = -0.3849 \text{ rad/s}^2,$$

As $\ddot{\theta}$ is negative $\Rightarrow \dot{\theta}$ is decreasing.

$$\therefore \vec{\alpha} = 0.3849 \hat{k} \text{ rad/s}^2 \text{ [Counter-clockwise]} \quad [3 \text{ Marks}]$$

$$\text{Now, } CA = [0.375^2 + 0.75^2] = 0.8385 \text{ m}$$

Tangential Acceleration

$$(\vec{a}_A)_t = CA \times \ddot{\theta} = 0.8385 \times 0.3849 \text{ m/s}^2 = 0.3227 \text{ m/s}^2 \text{ (Ans)} \quad [2 \text{ Marks}]$$

Normal Acceleration

$$(\vec{a}_A)_n = \omega^2 \times CA \text{ (From A to C)} = 1.1547^2 \times 0.8385 = 1.118 \text{ m/s}^2 \text{ (Ans)} \quad [2 \text{ Marks}]$$

Problem 4:

The cable from drum A turns the double wheel B, which rolls on its hubs without slipping. Determine the angular velocity ω and angular acceleration α of drum C for the instant when the angular velocity and angular acceleration of A are 5 rad/s and 4 rad/s^2 , respectively, both in the counter clockwise direction. Take $r_A = r_C = 0.1 \text{ m}$, $r_o = 0.4 \text{ m}$ and $r_i = 0.2 \text{ m}$. [10 Marks]

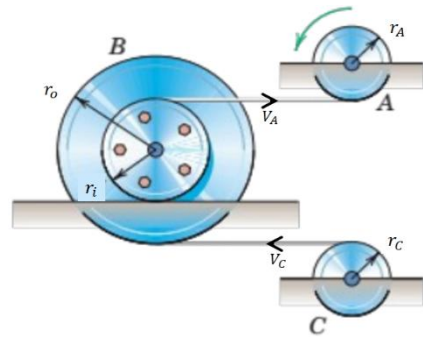


Figure 5: Problem 4

Solution:

$$\text{Given, } r_A = r_C = 0.1 \text{ m, } r_o = 0.4 \text{ m, } r_i = 0.2 \text{ m, } \omega_A = 5 \text{ rad/s, } \alpha_A = 4 \text{ rad/s}^2$$

$$\omega_A = 5 \text{ rad/s} \quad \curvearrowright$$

$$V_A = (0.1 \times 5) = 0.5 \text{ m/s} \quad \rightarrow$$

$$\omega_B = \frac{0.5}{0.4} \text{ rad/s} \quad \curvearrowright$$

$$V_C = \omega_B \times 0.2 \text{ m/s} = \frac{0.5}{0.4} \times 0.2 \text{ m/s} = \frac{5}{4} \times 0.2 \text{ m/s} = \frac{1}{4} \text{ m/s} \quad \leftarrow$$

$$\omega_C = \frac{V_C}{r_C} \text{ rad/s} = \frac{1}{0.1 \times 4} \text{ rad/s} = \frac{5}{2} \text{ rad/s} \text{ (Counterclockwise) (Ans)} \quad [5 \text{ Marks}]$$

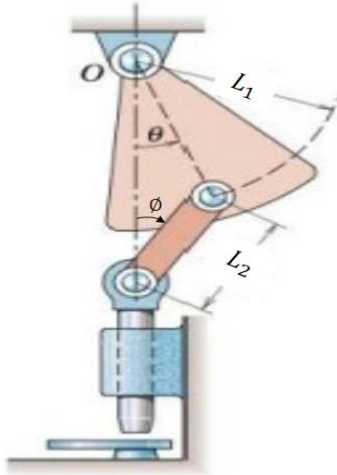
$$\alpha_A = 4 \text{ rad/s}^2 \quad \curvearrowright$$

$$a_{At} = 0.1 \times 4 \text{ m/s}^2 = 0.4 \text{ m/s}^2 \quad \rightarrow$$

$$\alpha_B = \frac{0.4}{0.4} \text{ rad/s}^2 = 1 \text{ rad/s}^2 \quad \curvearrowleft$$

$$a_{Ct} = \alpha_B \times 0.2 \text{ m/s}^2 = 0.2 \text{ m/s}^2 \quad \leftarrow$$

$$\alpha_C = \frac{a_{Ct}}{0.1} \text{ rad/s}^2 = \frac{0.2}{0.1} \text{ rad/s}^2 = 2 \text{ rad/s}^2 \quad (\text{Counter clockwise}) \quad (\text{Ans}) \quad [5 \text{ Marks}]$$



Problem 5:

The punch (Fig. 6) is operated by a simple harmonic oscillation of the pivoted sector given by $\theta = \theta_0 \sin 2\pi t$ where the amplitude is $\theta_0 = \pi/10$ rad (18°) and the time for one complete oscillation is 1 second. Determine the acceleration of the punch when (a) $\theta = 0$ rad and (b) $\theta = \frac{\pi}{12}$ rad, considering $L_1 = 300$ mm and $L_2 = 200$ mm. **[10 Marks]**

Solution:

Figure 6: Problem 5

Given, $L_1 = 300$ mm, $L_2 = 200$ mm

From above $\triangle OPQ$, $\frac{PQ}{\sin \theta} = \frac{OQ}{\sin \phi} \Rightarrow \sin \phi = \frac{3}{2} \sin \theta \quad \dots (1)$

Differentiating equation (1) w.r.t. time we get,

$$\cos \phi \dot{\phi} = \frac{3}{2} \cos \theta \dot{\theta} \Rightarrow \dot{\phi} = \frac{\frac{3}{2} \cos \theta \dot{\theta}}{\cos \phi} \quad \dots (2)$$

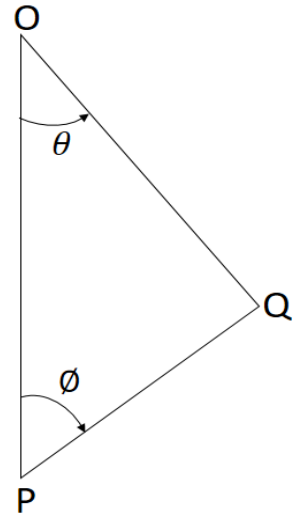
Differentiating equation (1) two times w.r.t. time we get,

$$\begin{aligned} -\sin \phi \dot{\phi}^2 + \cos \phi \ddot{\phi} &= -\frac{3}{2} \sin \theta \dot{\theta}^2 + \frac{3}{2} \cos \theta \ddot{\theta} \\ \Rightarrow \ddot{\phi} &= \frac{\frac{3}{2} \cos \theta \ddot{\theta} + \sin \phi \dot{\phi}^2 - \frac{3}{2} \sin \theta \dot{\theta}^2}{\cos \phi} \Rightarrow \ddot{\phi} = \frac{\frac{3}{2} \cos \theta \ddot{\theta} + \sin \phi (\dot{\phi}^2 - \dot{\theta}^2)}{\cos \phi} \quad \dots (3) \end{aligned}$$

Let, $OP = x = (0.3 \cos \theta + 0.2 \cos \phi) \text{ m} \quad \dots (4)$

Now, differentiating equation (4) w.r.t. time we get,

$$\dot{x} = -0.3 \sin \theta \cdot \dot{\theta} - 0.2 \sin \phi \cdot \dot{\phi} \quad \dots (5) \quad [2 \text{ Marks}]$$



Further differentiating equation (5) w.r.t. time we get,

$$\ddot{x} = -0.3 \cos \theta \cdot \dot{\theta}^2 - 0.3 \sin \theta \cdot \ddot{\theta} - 0.2 \cos \phi \dot{\phi}^2 - 0.2 \sin \phi \ddot{\phi} \quad \dots (6) \quad [1 \text{ Marks}]$$

$$\text{Now, } \theta = \theta_0 \sin 2\pi t \quad \dots (7)$$

$$\text{Then, } \dot{\theta} = \theta_0 \cos 2\pi t \cdot 2\pi, \quad \dots (8)$$

$$\text{and } \ddot{\theta} = -\theta_0 \sin 2\pi t \cdot 4\pi^2 = -4\pi^2 \theta \quad \dots (9) \quad [1 \text{ Marks}]$$

$$(a) \text{ When } \theta = 0, \phi = 0 \quad \therefore \sin \theta = 0, \sin \phi = 0, \text{ and } \cos \theta = 1, \cos \phi = 1$$

$$\text{From equation (7) } t = 0$$

$$\text{When } t = 0, \text{ from equation (8), } \dot{\theta} = \theta_0 \times 2\pi = \frac{\pi}{10} \times 2\pi = 1.974 \text{ rad/s}.$$

$$\text{When } \theta = 0, \text{ from equation (9), } \ddot{\theta} = 0.$$

$$\text{Now, from equation (2), } \dot{\phi} = 2.961 \text{ rad/s}$$

$$\text{From equation (3), } \ddot{\phi} = 0$$

$$\text{Then, finally from equation (6), } \ddot{x} = -0.3\dot{\theta}^2 - 0.2\dot{\phi}^2 = -2.923 \text{ m/s}^2$$

$$\therefore \text{ Acceleration of punch} = 2.923 \text{ m/s}^2 \text{ [up] (Ans)} \quad [3 \text{ Marks}]$$

$$(b) \text{ When } = \frac{\pi}{12}, \text{ from equation (1), } \phi = 0.3987 \text{ rad.}$$

$$\text{From equation (7) } t = 0.1568 \text{ sec,}$$

$$\text{When } t = 0.1568 \text{ sec, from equation (8), } \dot{\theta} = 1.091 \text{ rad/s}.$$

$$\text{When } \theta = \frac{\pi}{12}, \text{ from equation (9), } \ddot{\theta} = -10.335 \text{ rad/s}^2.$$

$$\text{Now, from equation (2), } \dot{\phi} = 1.7153 \text{ rad/s}$$

$$\text{From equation (3), } \ddot{\phi} = -15.5098 \text{ rad/s}^2.$$

$$\text{Then, finally from equation (6), } \ddot{x} = 1.1195 \text{ m/s}^2$$

$$\therefore \text{ Acceleration of punch} = 1.1195 \text{ m/s}^2 \text{ [down] (Ans)} \quad [3 \text{ Marks}]$$